Date:21/10/26

Experiment- 7

Continuous-Time Fourier Transform and Spectrum

OBJECTIVE:

(i) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.
(ii) To verify important Fourier transform properties and examine their effects on the corresponding time-domain and frequency-domain representations.

SOFTWARE / LABORATORY ENVIRONMENT:

MATLAB

I. To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

PROCEDURE:

1. Generate $x(t)=e^{(-t^2)}$.
2. Create $x(t-0.4)$ using interpolation.
3. Compute the FFT of the shifted signal.
4. Form the theoretical transform using the time-shift property.
5. Compare computed and theoretical magnitude and phase.

MATLAB CODE :

```
clc; clear; close all;

%figure;

% Magnitude and Phase Spectrum of a CT Signal

t = -2:0.001:2;

dt = t(2)-t(1);

x = exp(-t.^2).*cos(2*pi*3*t);

N = 4096;

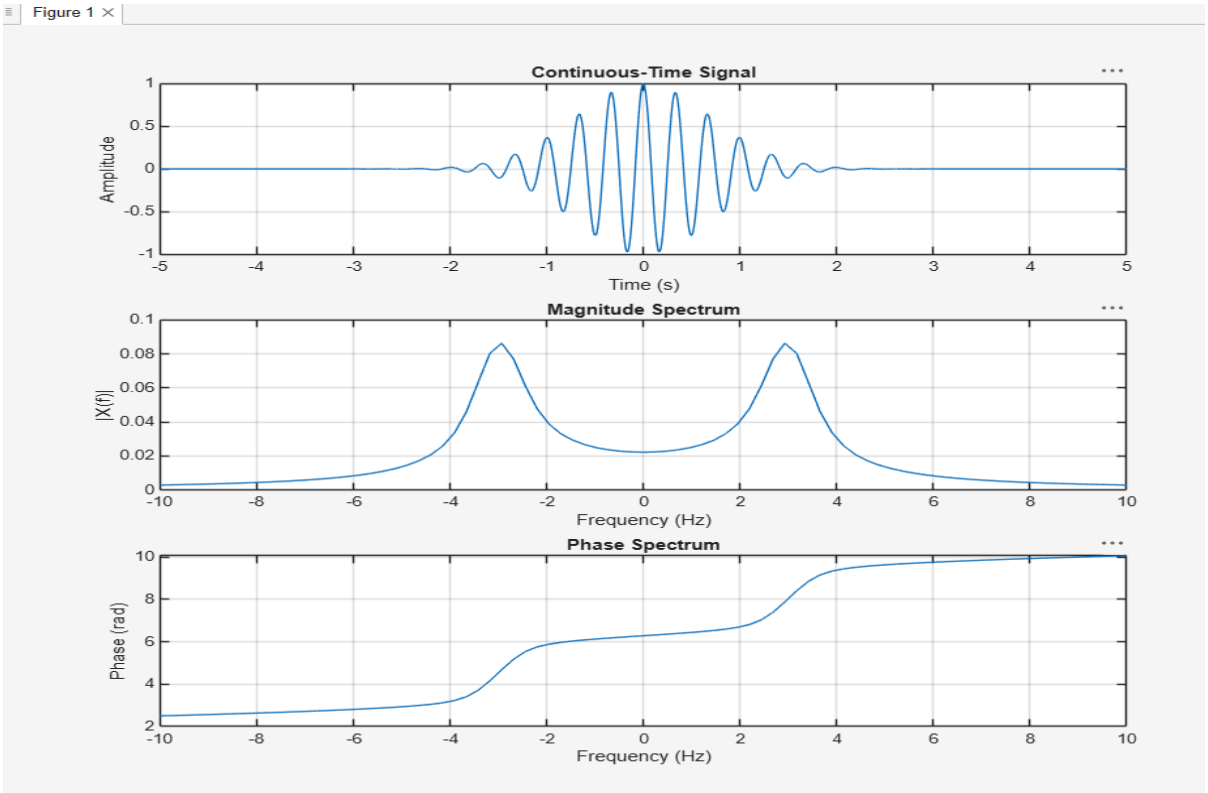
X = fftshift(fft(x,N))*dt;

f = (-N/2:N/2-1)/(N*dt);

magX = abs(X);
```

```
phaseX = unwrap(angle(X));
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Continuous-Time Signal');
subplot(3,1,2);
plot(f,magX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('|X(f)|');
title('Magnitude Spectrum');
subplot(3,1,3);
plot(f,phaseX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
title('Phase Spectrum');
```

OUTPUT:



II. To verify important Fourier transform properties and examine their effects on the corresponding time-domain and frequency-domain representations.

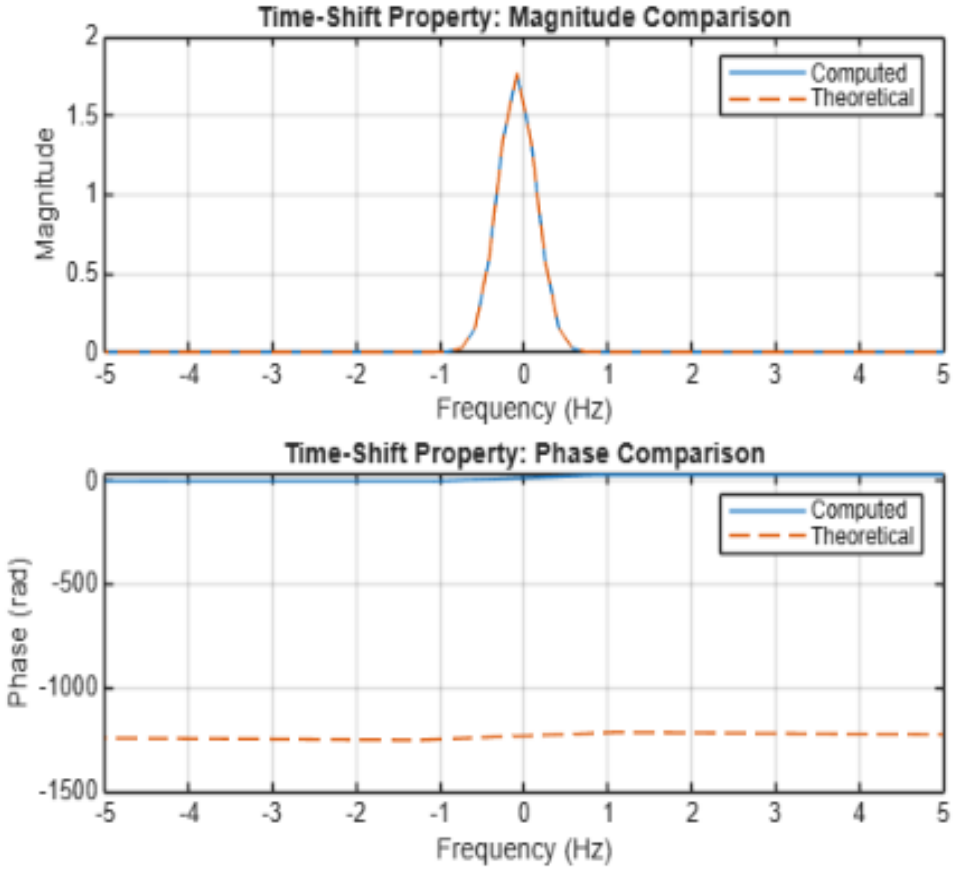
PROCEDURE:

1. Set fs = 2000 Hz and generate the composite signal.
2. Compute the FFT and shift the spectrum.
3. Plot the magnitude spectrum from −400 to 400 Hz.
4. Apply the 5% magnitude threshold.
5. Determine the approximate one-sided occupied bandwidth.

MATLAB CODE:

```
clc; clear; close all;
%figure;
% Verification of Fourier Transform Properties
t = -3:0.001:3;
dt = t(2)-t(1);
x = exp(-t.^2);
x_shift = interp1(t,x,t-0.4,'linear',0);
X = fftshift(fft(x))*dt;
N = length(x);
f = (-N/2:N/2-1)/(N*dt);
% Time-shift property: x(t-t0) <-> X(f)e^(-j2*pi*f*t0)
X_shift_theory = X.*exp(-1j*2*pi*f*0.4);
X_shift = fftshift(fft(x_shift))*dt;
subplot(2,1,1);
plot(f,abs(X_shift),f,abs(X_shift_theory),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');
subplot(2,1,2);
plot(f,unwrap(angle(X_shift)),f,unwrap(angle(X_shift_theory)),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
title('Time-Shift Property: Phase Comparison');
legend('Computed','Theoretical');
```

OUTPUT:



RESULT:

The continuous-time Fourier transform of representative signals was successfully obtained using FFT-based approximation, and their magnitude and phase spectra were analyzed to identify significant frequency components and verified important Fourier transform properties and examine their effects on the corresponding time-domain and frequency-domain representations.

LABORATORY PERFORMANCE ASSESSMENT RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.	**Fourier Transform Computation & MATLAB Implementation**	Computes X(jω) of every given CT signal correctly, analytically and numerically . Uses a suitable time range, sampling rate and frequency axis in rad/s or Hz.	Computes the spectra correctly, with minor errors in scaling, sampling rate, frequency axis or parameters.	Computes the spectra for only some signals, or has noticeable errors in code, scaling or frequency axis, and needs assistance.	Unable to compute the Fourier transform or produce meaningful spectra.	**20**	2	
	Magnitude & Phase Spectrum Plots	Plots $\|X(j\omega)\|$ and $\angle X(j\omega)$ correctly for every signal, with proper scaling, titles, axis labels and grid. Handles phase wrapping and phase near zero-magnitude points correctly.	Plots both spectra correctly, with minor errors in scaling, labelling, phase wrapping or axis range.	Plots the spectra with noticeable errors in scaling, phase handling or labelling, and needs guidance.	Unable to plot the magnitude and phase spectra.	**20**	2	

	Spectral Component Analysis	Correctly identifies dominant frequencies, peaks, bandwidth and the frequency range holding most of the signal energy. Uses symmetry (even magnitude, odd phase for real signals) and relates the spectral lines or lobes to the signal parameters.	Identifies most significant spectral components and bandwidth, with minor errors or limited justification.	Identifies only basic spectral components. Bandwidth or symmetry reasoning is incomplete, and needs guidance.	Unable to identify or justify the significant spectral components.	**20**	2	
	Analytical vs Numerical Verification	Compares the analytical and MATLAB spectra, quantifies the difference and explains its causes such as truncation, sampling and spectral leakage. Shows how signal parameters change	Compares analytical and numerical results correctly, with brief explanation of differences.	Makes a basic comparison with noticeable errors, and the explanation of differences is incomplete.	Unable to verify the numerical spectra against the analytical results.	**20**	2	

		the spectrum.						
	Engineering Analysis, Record & Interpretation	Provides technically accurate interpretation of the results and relates them to the physical or engineering meaning of the signal. Maintains a complete, well-organized record with observations, tables, plots and conclusions.	Maintains a proper record with correct interpretation, with minor omissions in explanation or conclusions.	Record and interpretation are partial or only descriptive, and need guidance.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	
					Total	**100**	**10**	